

Early response marker during pembrolizumab treatment in metastatic urothelial cancer: temporal shift in peripheral CD4 T cell subset expressing chemo-attractant receptors

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BACKGROUND

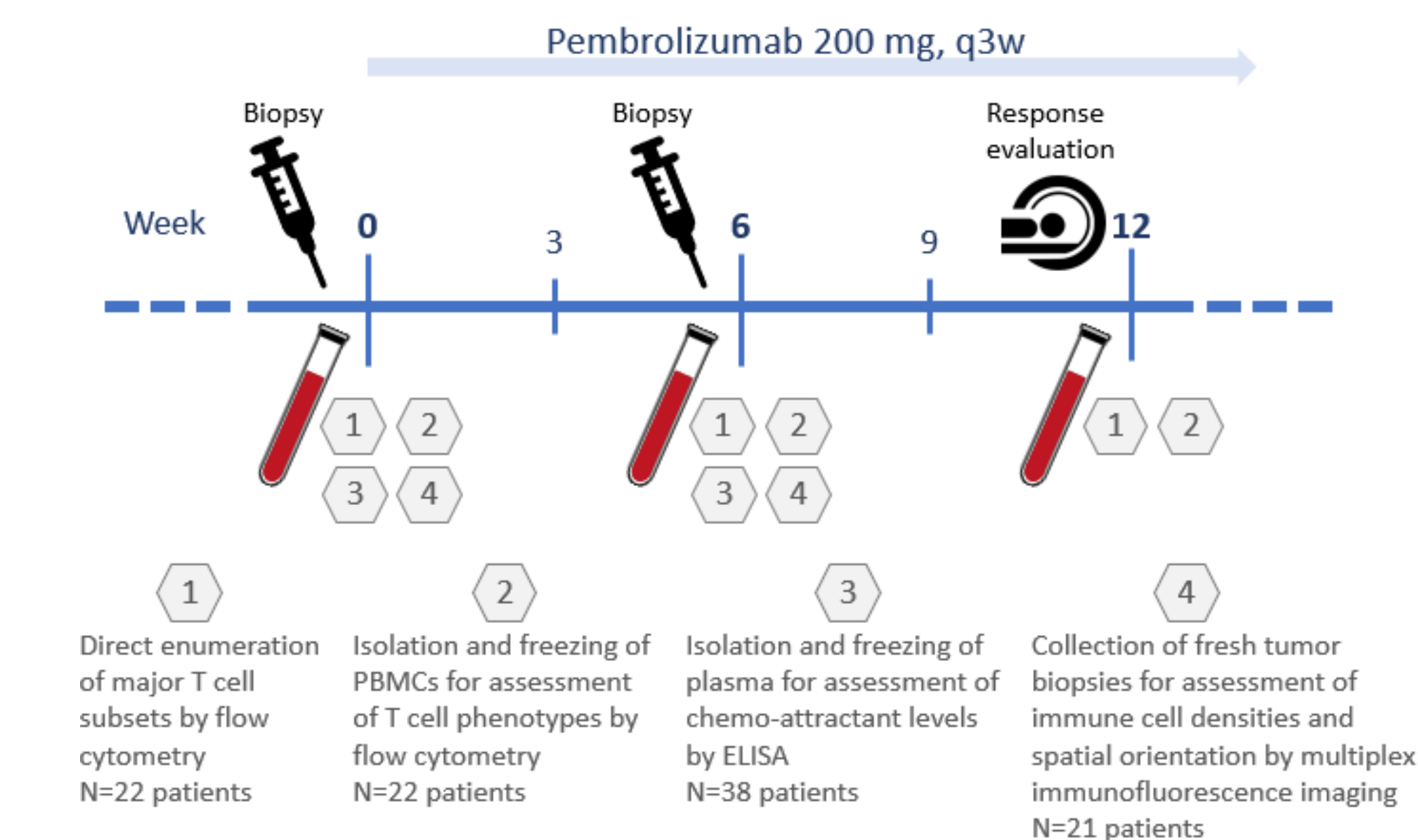
- Anti-PD1 has limited response rate in advanced and metastatic urothelial cancer
- PD-L1 is only predictive for response in first-line setting
- No predictive biomarkers are available for selection of patients for second-line treatment with anti-PD1

OBJECTIVES

- To identify early biomarkers for clinical benefit during pembrolizumab treatment
- To identify the mechanism of action of primary and acquired resistance to pembrolizumab

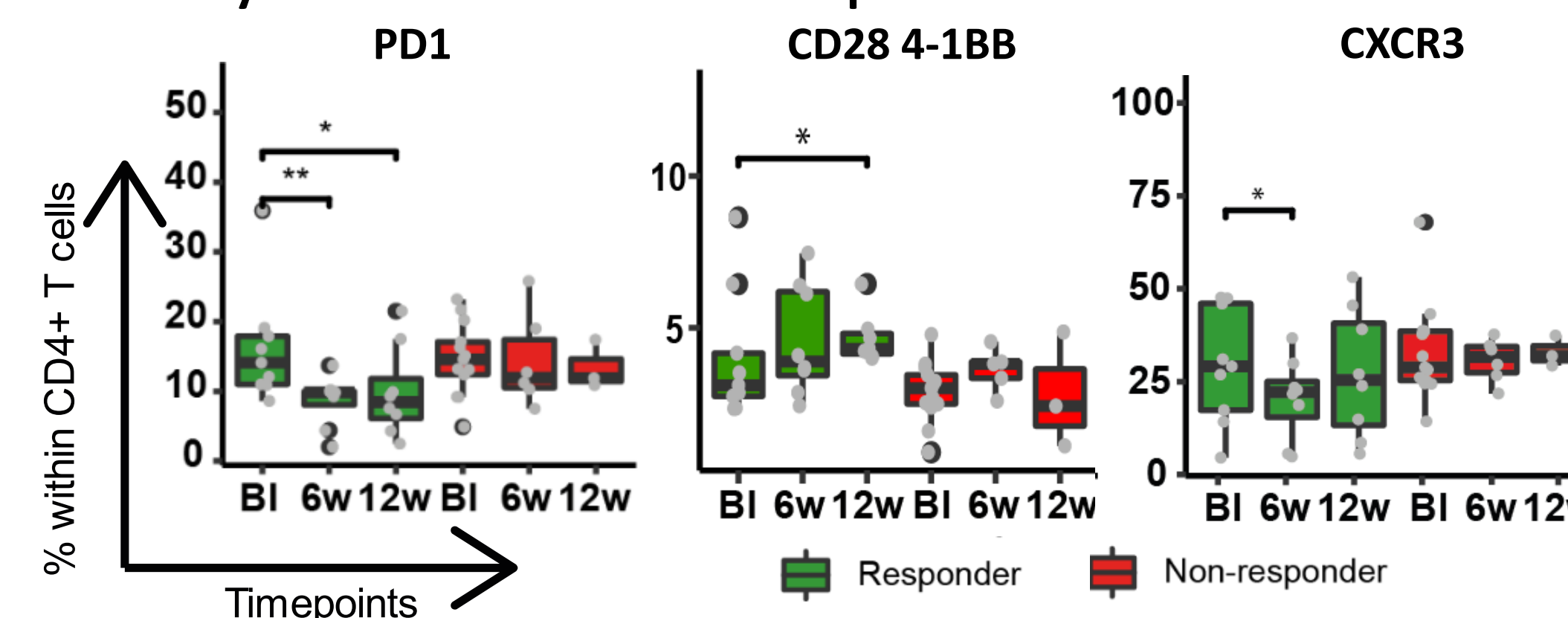
PATIENTS AND METHODS

- Patients with advanced or metastatic urothelial cancer
- Patients received no prior systemic chemotherapy, or had progressive disease after platinum containing chemotherapy

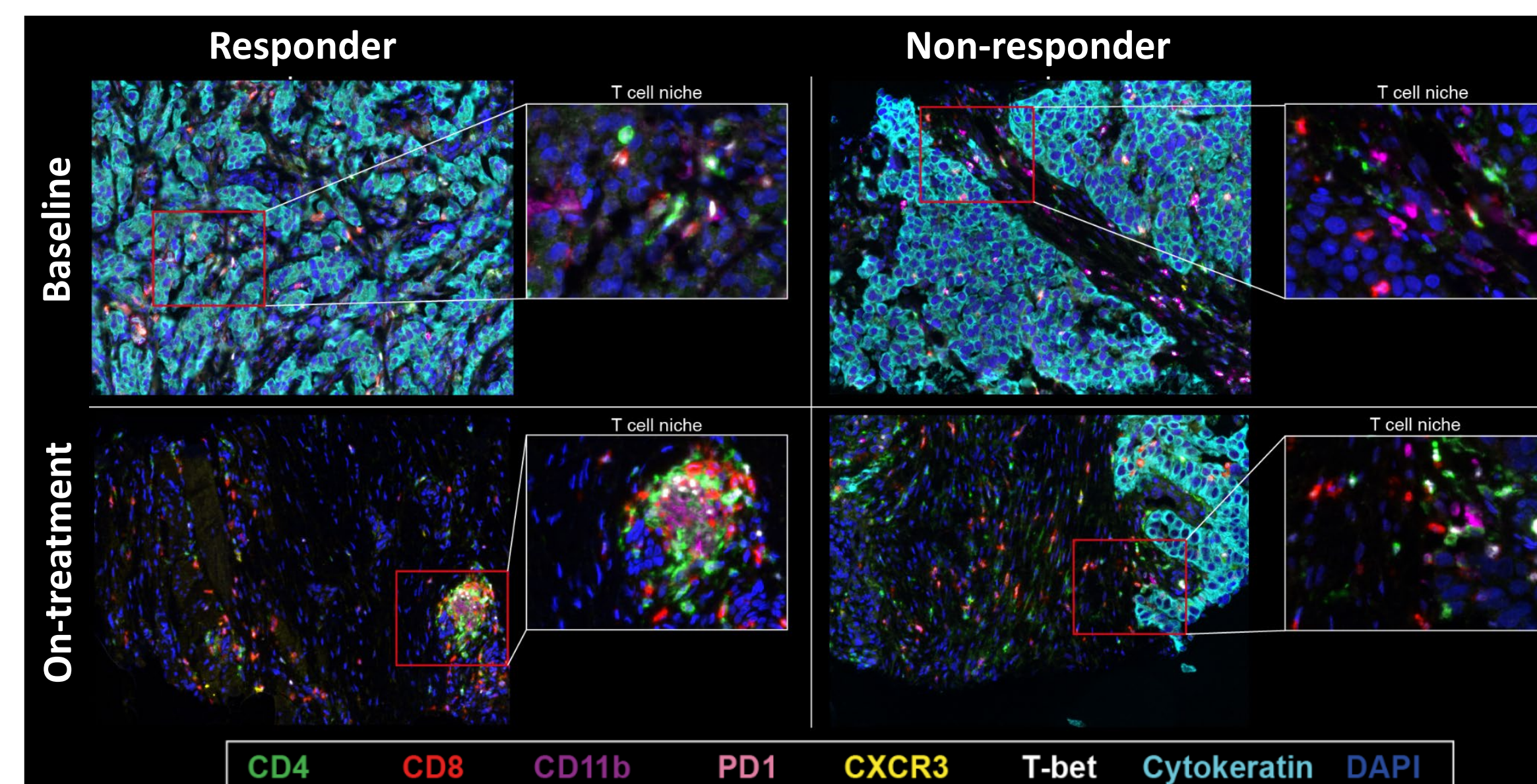


RESULTS

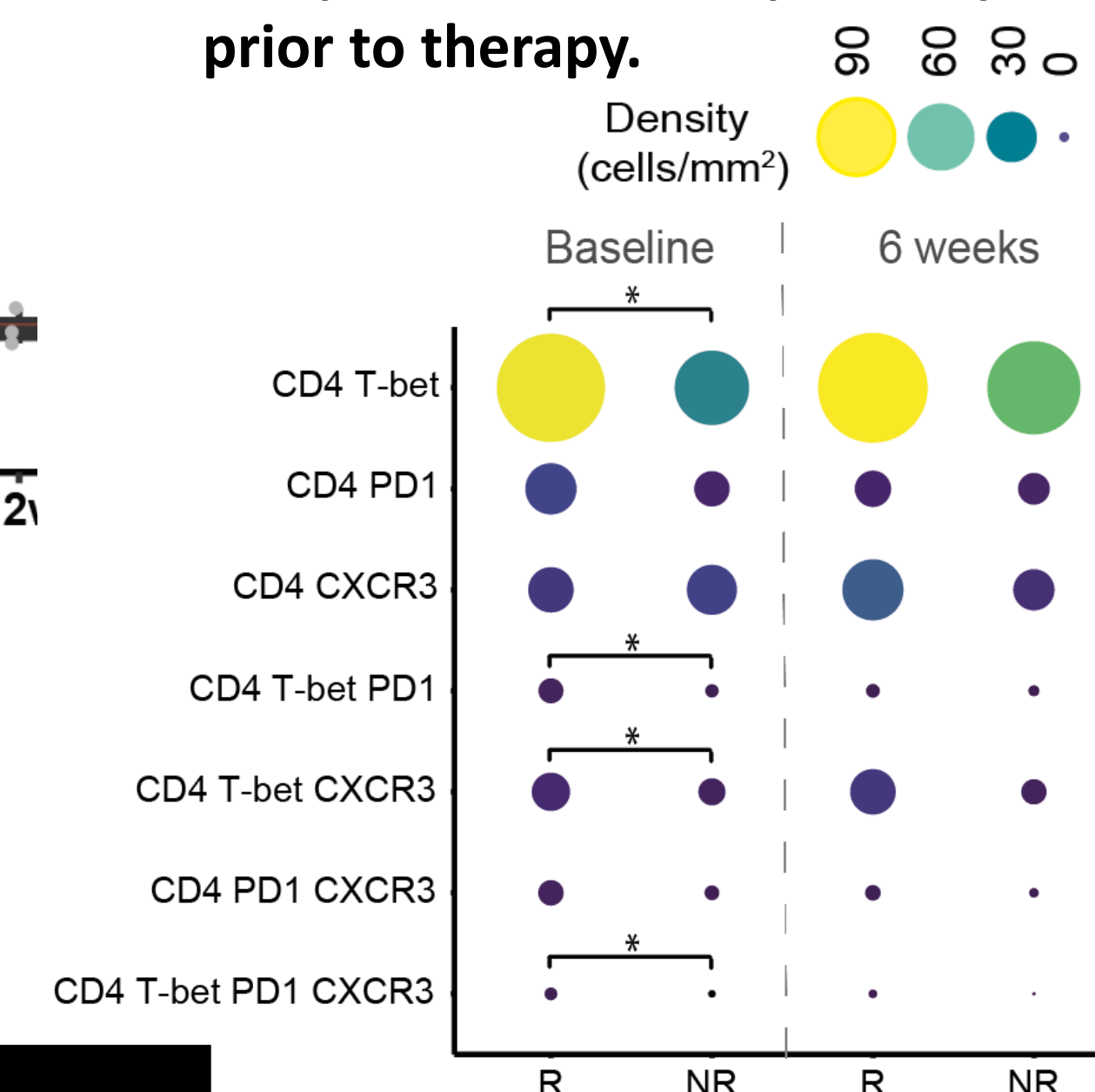
CD4 T cell subsets in blood according to expression of co-inhibitory, co-stimulatory or chemo-attractant receptors.



Representative multiplex immunofluorescence images of tumor tissues obtained from a responder (abdominal wall metastasis) and non-responder (intra-abdominal metastasis) patient at baseline and after 6 weeks of treatment.

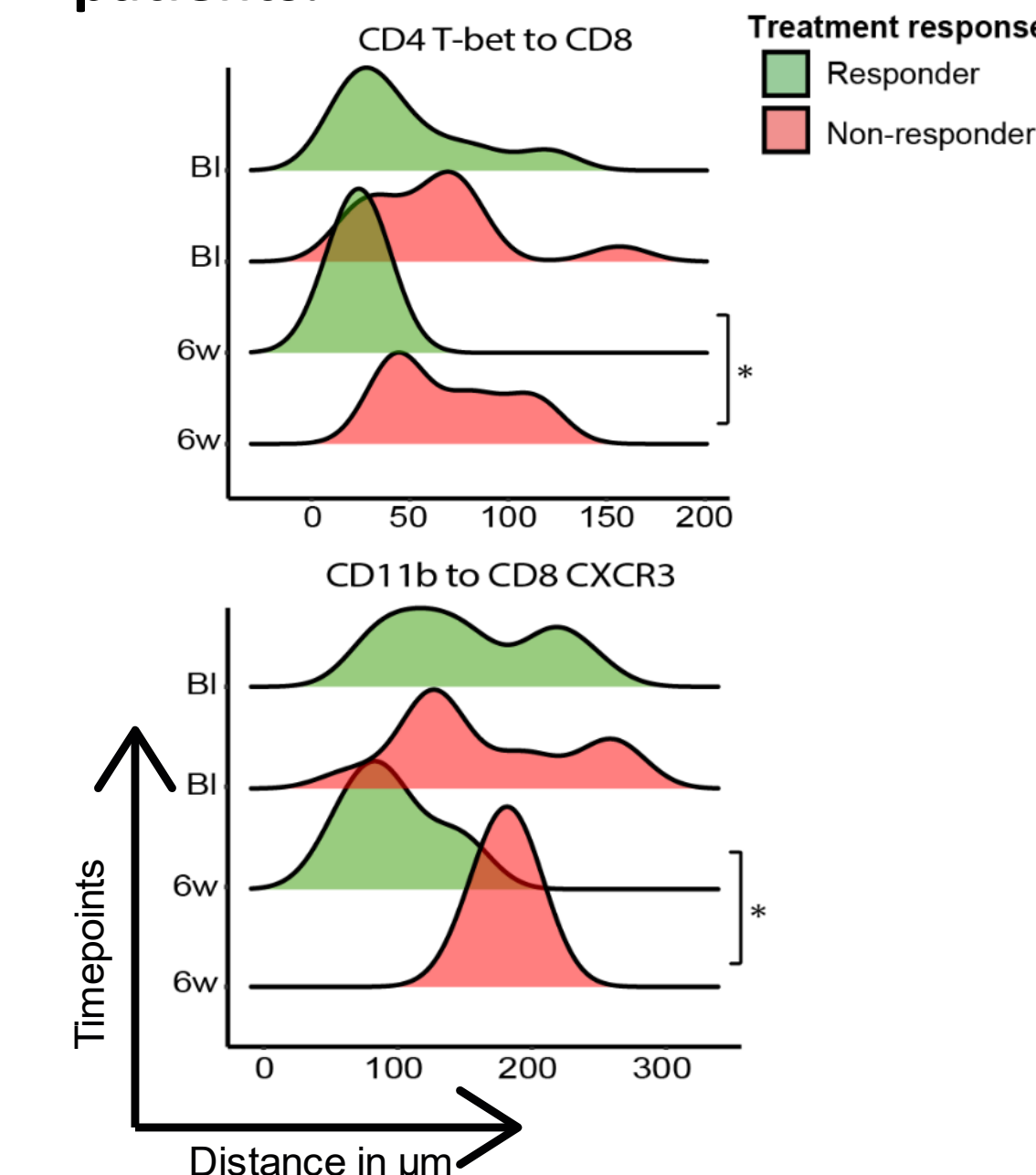


Intra-tumoral density of T-bet+ CD4+ T cells is higher in responder when compared to non-responder patients prior to therapy.



Circle plots show mean densities (cells/mm²) of CD4+ subsets according to T-bet, PD1 and/or CXCR3 expressions.

Intra-tumoral T cells and myeloid cells form immune cell niches during treatment in responder patients.



Histograms show mean distance in μm (x-axis) from CD4 T-bet+ to CD8+ T cells, and CD11b+ to CD8 CXCR3+ T cells versus cell densities (cells/mm², y-axis).

CONCLUSION

In patients with metastatic urothelial cancer, favorable tumor response to anti-PD1 can be predicted by systemic changes in chemo-attractant receptor-expressing CD4+ T cells together with formation of intra-tumoral niches that are rich in immune effector cells.